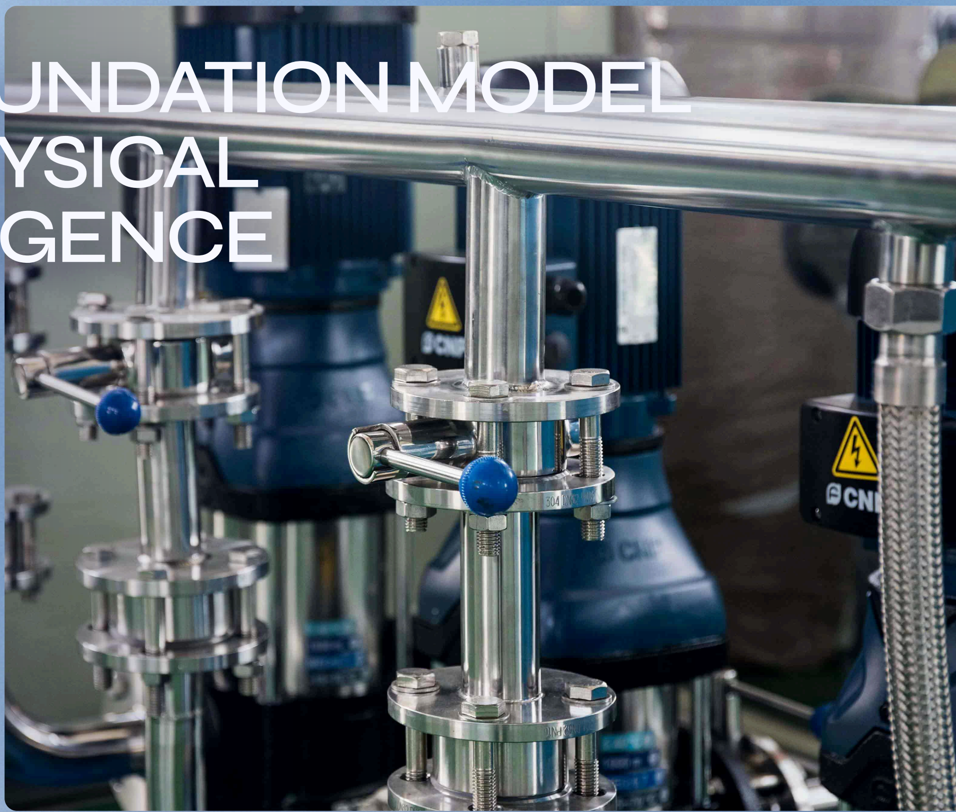
CONFIDENTIAL

SHODH

THE FOUNDATION MODEL FOR PHYSICAL INTELLIGENCE



From molecule to factory

CONFIDENTIAL INVESTMENT MEMORANDUM

$55M EQUITY FINANCING

JULY 2026

In recent prospective industrial validations, Shodh's uniform physics model compressed a 3-month physical formulation test into 24 hours of computation for a global CRO, and successfully improved chemical factory yield from 82.4% to 96.7% on a mathematically locked run.

UNIFORM PHYSICS WORLD MODEL

MOLECULE → FACTORY

01 / INVESTMENT MEMORANDUM

Executive Summary

01

THE INDUSTRIALISATION GAP

A common intelligence layer for the physical economy.

While billions of dollars have been invested in "Discovery AI" to invent new digital molecules, and "Engineering AI" to optimise finished downstream machines, the physical economy remains bottlenecked by a massive, multi-trillion-dollar void in the middle: the **Industrialisation Gap**. Translating a digital molecular discovery into a mass-produced physical reality requires surviving formulation stability, chemical route redesign, and factory scale-up. Historically, this entire lifecycle has been a fragmented sequence of expensive, manual physical trial and error.

Shodh is a frontier foundation model for physical intelligence built to eliminate this bottleneck. Rather than building bespoke simulations for individual domains, Shodh operates a **Uniform Physics World Model** capable of true multi-scale, cross-domain inverse design. Whether calculating the thermodynamic degradation of a molecule or the fluid dynamics of a 10,000-litre factory vessel, Shodh computationally generates the physical parameters required to achieve a targeted industrial outcome.

TO COMMERCIALISE THIS INTELLIGENCE LAYER, SHODH DELIVERS THE MODEL THROUGH TWO ENTERPRISE MODULES:

Having validated the platform across multiple physical scales, and having already secured paid pilots structured for automated conversion into six-figure enterprise SaaS licenses, Shodh is raising **$55M in equity capital**. This round funds the compute required to ride our empirically proven scaling laws, transitioning the company from a proven scientific breakthrough into the dominant intelligence platform for the physical economy.

SHODH SYNTH / PROCESS INTELLIGENCE

SHODH SYNTH

Shodh SYNTH: The step-change upside. A process-invention capability that computationally redesigns entire chemical and biological manufacturing routes, yielding massive milestone payouts and selective process-IP royalties.

SHODH SCALE / FACTORY INTELLIGENCE

SHODH SCALE

Shodh SCALE: The recurring-revenue foundation. A factory and formulation deployment that optimises existing processes, capturing a "SaaS floor" base license (e.g., $250K–$500K/year) plus a capped gain-share on the verified economic value created.

ONE PLATFORM · TWO ENTERPRISE MODULES

INVESTMENT CASE AT A GLANCE

Proven today. Built to generalise.

Proven Today

Multi-Scale Physical Results:
Prospective factory scale-up (Aarti Industries: +14.3 pts in yield) and zero-day formulation prediction (Syngene: matching 3-month wet-lab data in 24 hours).

Commercial Traction:
Paid validations actively converting to $250,000/year recurring enterprise site licenses.

Frontier Technology:
Core Uniform Physics World Model built, with proven empirical power-law scaling capability.

Customer Pull:
9 active/scoped programmes across pharma, specialty chemicals, biologics, and aerospace.

What This Round Establishes

Generalisation:
Repeatability across expanded process classes via dozens of parallel physical validations.

Capability Unlocks:
Scaling H100/B200 compute infrastructure to enable zero-shot transfer across unseen domains.

Software Margins:
Declining human adaptation effort and computational hours per new programme.

Commercial Scale:
Repeatable private SCALE deployments and a scalable organisation with dedicated commercial leadership.

2. The Industrial Scale-Up Gap

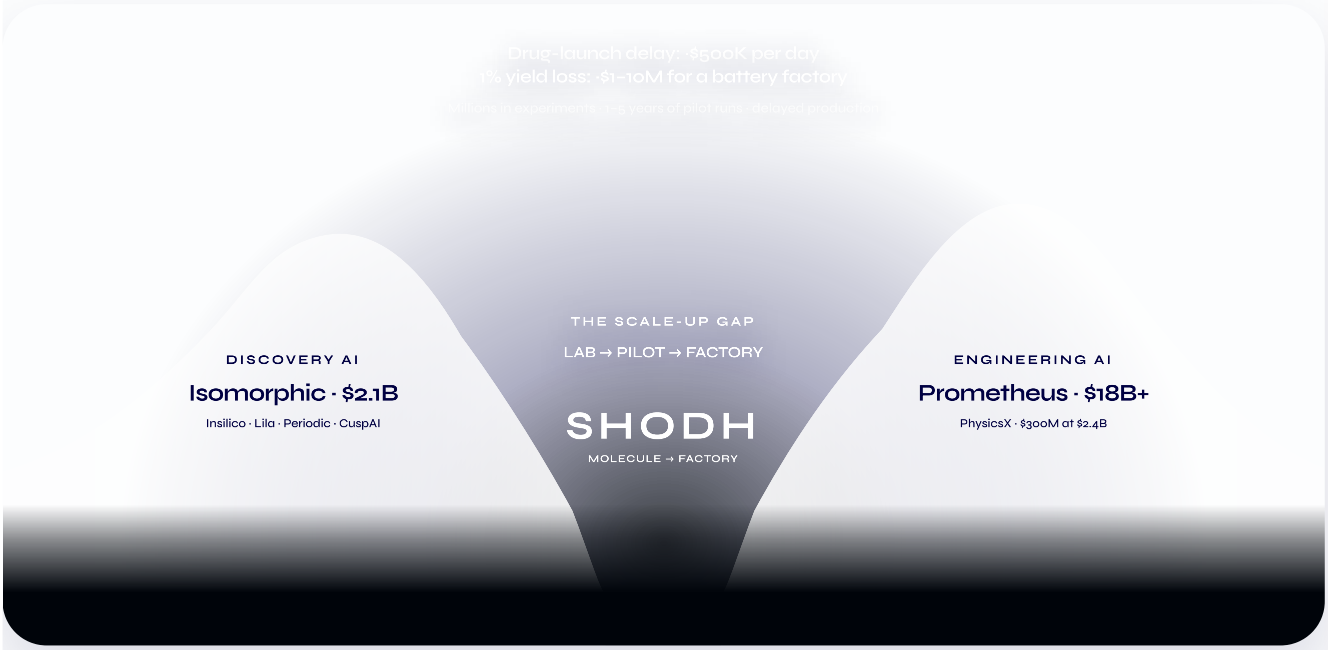
The greatest unsolved bottleneck in the physical sciences is moving from discovery to manufacturing.

On one side, Discovery AI companies—including Isomorphic, Insilico, Lila, Periodic and CuspAI—have attracted billions of dollars to design molecules and materials at the microscopic level. On the other, Engineering AI is advancing the design of large-scale machines and hardware, illustrated by PhysicsX's $300M raise at a reported $2.4B valuation and the more than $18B committed to Prometheus's ambition of building an "artificial general engineer" for systems such as jet engines and robotics.[1–7]

Between these two categories remains a structural valley of death:

> How does a molecule that works in the laboratory become a reliable, economical industrial process?

This is the scale-up gap.



A laboratory breakthrough cannot simply be enlarged into a factory. When a reaction, biological process or material moves from a benchtop system into a 10,000-litre vessel, its physical environment changes. Heat no longer dissipates in the same way. Mixing becomes non-uniform. Mass transfer, shear, residence time, fluid dynamics, reaction kinetics and equipment geometry begin interacting at industrial scale.

Although the specific failure modes differ, the structure of the scale-up problem repeats across industries. In chemicals, the constraint may be heat release, mixing and impurity formation; in biologics, shear, mass transfer and aggregation; in batteries and advanced materials, phase behaviour, coating uniformity and thermal control. The domains differ, but the underlying challenge is the same: molecular behaviour must be translated through transport physics and equipment constraints into a stable industrial process.

This recurring physical structure is what makes scale-up potentially addressable through a common intelligence platform rather than a collection of unrelated engineering projects.

The resulting process knowledge is among the most valuable and closely guarded forms of industrial intelligence. TSMC's dominance was not created only by knowing what semiconductor to manufacture; it was created through decades of accumulated knowledge about how to manufacture at exceptional yield and reliability. Similar process advantages determine competitiveness across pharmaceuticals, chemicals, biologics, batteries and advanced materials.

The limitation of the existing stack

Manufacturers currently manage scale-up through experienced process teams, deterministic simulation, specialist consultants and successive pilot runs.

These systems are powerful, but they largely operate in the forward direction: engineers specify a proposed process, equipment configuration and set of operating conditions, and the software evaluates what may happen.

The harder problem is the inverse:

> Given a required yield, impurity threshold, cost and production scale, what process, equipment configuration and operating conditions should be used?

The possible design space can contain thousands of interacting molecular, thermodynamic, transport and equipment variables. Even the most experienced engineering teams must reduce this space through assumptions, isolated simulations and physical trial and error. The result is a long sequence of laboratory experiments, pilot campaigns and process revisions before reliable production can begin.

The economic penalty is severe:

Pharmaceuticals: a delayed drug launch can destroy more than $500,000 of revenue per day.

Energy and materials: a 1% yield loss in a battery factory can represent 1 M – 1M– 10M of annual economic loss.

Process industries: unsuccessful scale transfer can consume years of engineering work, physical experimentation and delayed production.

These are not software-efficiency problems. They are physical-production problems measured in lost output, wasted material, delayed revenue and stranded industrial capacity.

The inverse-design opportunity

The missing layer is an intelligence system that can connect molecular behaviour, process conditions and equipment-scale physics—and work backwards from the required industrial outcome.

Shodh is being built to provide that layer.

Given the molecule or biologic, laboratory and process data, equipment geometry and operating constraints, Shodh predicts industrial outcomes and generates the process conditions required to achieve a defined manufacturing target. Instead of asking engineers to manually search the complete design space, Shodh uses a common physical-intelligence model to perform inverse design across molecule, process and equipment.

Discovery AI determines what might work. Engineering software evaluates designs that have already been specified. Shodh determines how a scientific result can become a manufacturable industrial process.

By converting scale-up from a fragmented sequence of expert judgement, isolated simulation and physical trial and error into a reusable intelligence layer, Shodh is targeting one of the largest remaining bottlenecks in the physical economy.